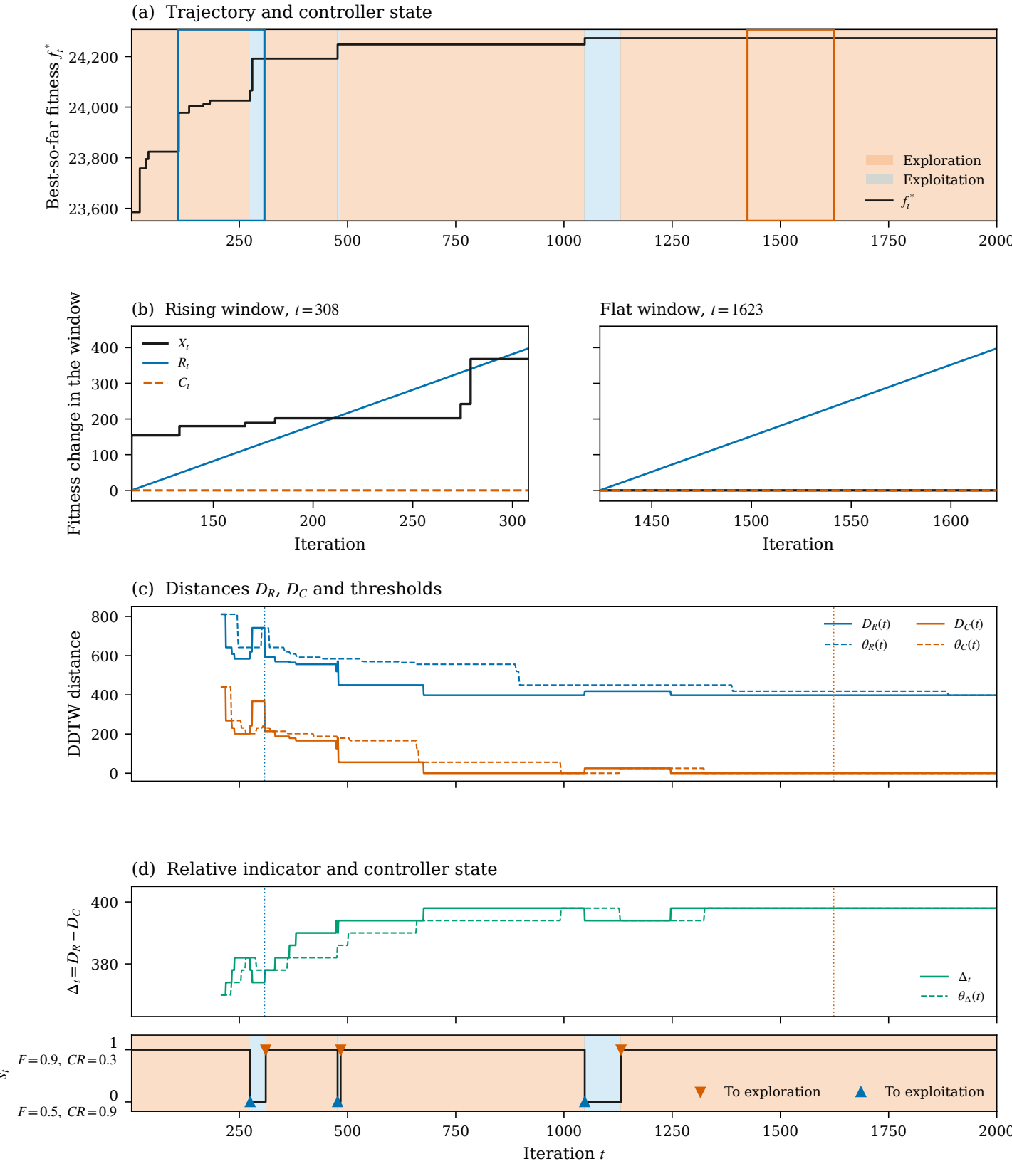
(a) Trajectory and controller state
Best-so-far fitness f_t^*
24,200
24,000
23,800
23,600
250
500
750
1000
1250
1500
1750
2000
Exploration
Exploitation
f_t^*
(b) Rising window, $t = 308$
Flat window, $t = 1623$
Fitness change in the window
400
300
200
100
0
X_t
R_t
C_t
150
200
250
300
1450
1500
1550
1600
Iteration
Iteration
(c) Distances D_R, D_C and thresholds
DDTW distance
800
600
400
200
0
$D_R(t)$
$\theta_R(t)$
$D_C(t)$
$\theta_C(t)$
(d) Relative indicator and controller state
$\Delta_t = D_R - D_C$
400
380
Δ_t
$\theta_\Delta(t)$
s_t
1
0
$F = 0.9,\ CR = 0.3$
$F = 0.5,\ CR = 0.9$
To exploration
To exploitation
250
500
750
1000
1250
1500
1750
2000
Iteration t